

Data Exchange Framework



Release Notes

Product Version: v12.5.2.1
Publication Date: March 2026

TABLE OF CONTENTS

Revision History	10
Release v12.5.2.1 (March 18, 2026)	11
Enhancements	11
Improved the speed of the Workflow Enlighten action.....	11
Remove the default Email address from the NEAT Notification.	11
Support to ingest the data that is related to transferring to virtual agents.	11
DEF returns the proper error message when JSON doesn't return any participants.	11
Mandatory to use the LanguageID configuration in DEF action.	11
Enhance the way DEF handles the complex LLM prompt.....	11
DEF will calculate sentiment by default rather than NIA.....	12
New Actions / Features	12
Extract the complaints data from TAS 5.5 and send them to the NIA.....	12
Google Metadata Media Retrieve Action - Integration with Google API to retrieve media and metadata.....	12
Generate Desktop Tags Action - Supports ingesting 3rd party evaluation and converting them into desktop tags.	12
Merge two DEF actions into one action.	12
Key Resolved Issues	Error! Bookmark not defined.
Release v12.5.2 (January 26, 2026)	13
General Updates	13
Warning	13
New Features	13
Ability to integrate DEF with the v11 VoiceGrid.	13
Ability to integrate DEF with the v11 ASR engine in Analytics Hub.	13
Supports parsing metadata from transcripts generated by real-time ElevateAI.....	14
Retrieves the IPM model file from the AH to generate Topic AI data.....	14
Rename two DEF actions to Topic AI Request and Topic AI Response.....	14

Adds the ability to write to an S3 resource with and without a KMS key.....	14
Supports converting the v11 ASR transcript to ASR format.....	14
Added support to generate ENL scores for the v11 transcripts.....	14
Updated the way DEF calculates the best phrases.	14
Installation/CU updates & enhancements	15
An ability to view the history of the patches that have been applied.	15
DEF New Actions and Resources	15
AWS SQS resource	15
Analytics Hub Request	15
Analytics Hub Response:	15
Elevate AI Near Real-Time Retriever.....	15
Compatibility Matrix	16
Key Resolved Issues	Error! Bookmark not defined.
Known Issues/Behaviors and Workarounds	17
Release v12.5.1 (July 18, 2025)	18
General Updates	18
Warning	18
New Features	18
Provide the ability to decrypt the password stored in the DEF action.	18
Provide the ability to automate key rotation without the Assumed Roles.	18
Ability to differentiate various DEF instances for monitoring purposes.....	18
Ability to collect some additional metrics for the chat media for the new ingest path.	18
The DEF's enhanced logs show information from only one process at a time.	19
Provide a mechanism to identify whether the CSV file is complete or not.	19
Allows skipping the bad CSV file during DEF processing.....	19
Allows the option to rename fields to avoid conflicts.....	19
The sensitive fields were removed from the Call Summarization Response.	19
Enhancements to the "Cleanup Type" action for better folder management.	19
Provide the ability to generate an ASR format transcript from emails.	19

Provide the ability to generate an ASR format transcript from general text media types.	20
DEF shows a warning if there are mappings that are not in the working table.....	20
Allow Process Launch and Process Chain actions to refer to the launched process via ID.	20
Need to empty the key fields while importing/exporting processes.....	20
Associate the right speaker with the right channel.....	20
Add information such as scheduling, threads, and resources to the DEF Templates.	21
Enhanced CXone Media Retriever action to support IVR Data.....	21
Support for the SOCKS5 Proxy was added to the SFTP resource.	21
Pass the "SentimentTransition" value from the VG to the NIA.	21
Provide a separate retention period at the process level.....	21
The CXone Users Retriever v2 Action can return "hireDate."	21
Optimize the AppLink Metadata Retriever to process a large number of data.	21
The CXone Metadata Retriever skips records with a null ACD ID.	22
DEF captures additional info in the Process snapshots.	22
DEF adds the middle name to the user details CSV.....	22
Provide the ability to redact the email media type.	22
Sends a few missing data to the NIA first-class fields for the Email media types.....	22
Provides the ability to trace the interaction E2E between DEF and NIA.	22
Simplify the failed interactions re-processing task.	22
A few enhancements to the AutoSummary feature. All these changes are part of the Call Summarization Response v2 action.	22
Installation/CU updates & enhancements	23
Updated the configuration file to work with the NxDeployer to automate the DEF upgrade process.	23
Move the Workbench folder from the data folder into the install folder.	23
Updated the support for Java version from 17 to 21.....	23
Support for Windows Server 2022.	23
DEF New Actions and Resources	24
Allow the extraction of DFO emails from the CXone for the Nexidia Analytics.	24
Ability to retrieve data using Zoom API.	24

Provide the ability to feed billing data to the CXone billing mechanism.	24
Added a new resource type - Snowflake	24
Added a new resource type - Artemis.....	24
Allows the submission of a query to Snowflake via the Artemis message broker.	24
Allows the extraction of metadata from Snowflake using a query from a message broker...	25
Compatibility.....	25
Key Resolved Issues	26
Known Issues/Behaviors and Workarounds	27
Release v12.5.0.2 (February 28, 2025)	28
General Updates	28
Create Workflow Enlighten for SentimentValue and First Class Field	28
DEF's Inbox Publisher test mode supports the Extractor metadata format.	28
DEF's Inbox Publisher test mode supports re-processing.	28
DEF shows the older release name on the login page.	28
Send the Sentiment value information to the NIA's first-class field.	28
DEF's release artifact's Zip filenames reflect release name.....	28
Enhancements to the DFO chat integration.	29
Key Resolved Issues	29
Release v12.5.0.1 (January 31, 2025)	30
General Updates	30
Enhancements to the AutoSummary features.	30
Enhancements to Salesforce chat integration to remove chatbot data.	30
Enhancements to DFO chat integration.	30
Help migrate existing customers to Pre-ingest ASR.	31
Provide an ability to retrieve data using Zoom API.....	31
Allows to extract DFO Emails from the CXone APIs.	31
Enhancing Twilio API integration action.	31
Collects additional metrics for the chat media for the new ingest path.	31
Release v12.5 (December 20, 2024)	32

General Updates	32
Breaking Changes	32
New Features.....	33
Updated DEF's self-signed certificate that allows to access NEAT UI over HTTPS.	33
Ability to support different intent models in the new VG cluster arrangement.	33
Provide an ability to hide/show DEF features based on certain configurations.	33
Provide an ability to select media based on the metadata selection.	33
Make the deployment cluster-aware using the heartbeat.....	33
Provides the IPM information to the NIA to retrieve them later for other use.....	34
Updating the DEF monitoring data to support UTC format.....	34
Provide an ability to recalculate the sentiment and ENL scores without re-ASR.....	34
Update the CXone Metadata Retriever action based on the CXone API changes.	34
A couple of minor enhancements were added to the rule-based summary.	34
DEF now provides the top 3 intents of the interactions.....	35
Extract a few more fields from the Engage for the AppLink.	35
Add "publish confirmations" to both DEF and VG clients to confirm that processing is done.	35
Enhance Enlighten scores functionality to be compatible with ElevateAI integration.	35
Provide an ability to automate key rotation with the Assumed Roles.	35
Ability to use a specific intent model based on the interactions metadata.	35
Allow users to resize windows in the DEF UI using a mouse.....	36
Replace the custom SQL that generates the index scores for the AutoSummarization tag. ...	36
Ability to create behavior scores metadata in the Workflow Enlighten action.	36
Create _Thresholds columns for the Enlighten models when there are values in the threshold fields.	36
Provide a way for the FileDiscovery to work with other columns too besides "FileName."	36
Deletes the "*.ASR.json" files after processing from the Call Summarization Request.	37
DEF ensures chat durations are not zero.....	37
Added a min/max time option to resolve the issue around retrieving decryption keys.	37
Updated default query in AppLink Metadata Retriever to avoid purged calls.....	37

Provides an ability to limits the number of events to be displayed.	37
Provides the ability to remove existing custom actions and better validations.	38
Enhancements to the Enlighten processing.	38
Allow the use of newly created NIA first-class fields without upgrading the DEF.	38
Updated the default value for "Download voice and screen media" to false in the CXOne Media Retriever V2 action.....	38
Ignore the pair when the property name is empty for CXOneGenericMetadataRetrieverV2.	38
Allows to enter negative values in the Quantum field.....	38
Provide the ability not to publish the copied process automatically.	39
Enhance the "Remove Duplicate Metadata" action.	39
DEF allows an option to use HTTPS in agent/user communication to NIA.	39
DEF displays the release name, e.g. 12.4.1FP2, rather than the release number, e.g. 12.4.1.152.	39
Provide an ability to optionally check signing before decrypting in GPG Decrypt action.....	39
Provide the ability to choose either an interval or fixed execution schedule.	39
Provide the ability to skip bad emails and let the process continue to other media.....	40
Enhance Genesys Cloud Job Metadata Retriever Action.	40
Enhance dealing with "Forbidden" responses from Genesys Cloud Call and Recording Metadata Retriever Actions.	40
Handles the intermittent issue with CTI data on the first of the month.	40
Created a utility package that can be patched alongside actions.....	40
Installation/CU updates & enhancements.....	41
Improve automated deployment of the DEF using the MultiInstall file.	41
The default DEF ports are changed, and the user can edit them.	41
Better management of the existing patches.	41
Adds support for MSSQL 2022.....	41
Provide an ability to track custom actions better.	41
DEF New Actions and Resources.....	42
AutoSummary LLM	42
AppLink SAP Generic Upload	42

Call Summarization Request v2 AND Call Summarization Response v2.....	42
Calabrio User Metadata Retriever	42
Salesforce Metadata Parser	42
Salesforce Metadata Retriever.....	43
Convert Transcript to ASR.....	43
Custom SQL V2.....	43
Compatibility.....	43
Key Resolved Issues	44
Known Issues/Behaviors and Workarounds	45

COPYRIGHT

PROPRIETARY AND CONFIDENTIAL INFORMATION

Information herein is proprietary information and trade secrets of NiCE Ltd. and/or its affiliated companies (Affiliates). This document and the information herein is the exclusive property of NiCE and its Affiliates and shall not be disclosed, in whole or in part, to any third party or utilized for any purpose other than the express purpose for which it has been provided.

IMPORTANT NOTICE

Subject always to any existing terms and conditions agreed between you and NiCE or any Affiliate with respect to the products which are the subject matter of this document, neither NiCE nor any of its Affiliates shall bear any responsibility or liability to a client or to any person or entity with respect to liability, loss or damage caused or alleged to be caused directly or indirectly by any product supplied or any reliance placed on the content of this document. This includes, but is not limited to, any interruption of service, loss of business or anticipatory profits or consequential damage resulting from the use or operation of any products supplied or the content of this document. Information in this document is subject to change without notice and does not represent a commitment on the part of NiCE or any Affiliate.

All information included in this document, such as text, graphics, photos, logos, and images, is the exclusive property of NiCE or its Affiliate and is protected by United States and international copyright laws. Permission is granted to use, view and photocopy (or print) materials from this document only in connection with the products to which this document relates and subject to the terms of license applicable to such products. Any other use, copying, distribution, retransmission or modification of the information in this document without the express prior written permission of NiCE or an Affiliate is strictly prohibited. In the event of any permitted copying, redistribution or publication of copyrighted material, no changes in, or deletion of, author attribution, trademark legend or copyright notice shall be made.

Products supplied may be protected by one or more of the US patents listed at www.nice.com/Patents.

For the full list of trademarks of NiCE and its Affiliates, visit www.nice.com/NiCE-Trademarks. All other marks used are the property of their respective proprietors.

All contents of this document are: Copyright © 2025 NiCE Ltd. All rights reserved.

Revision History

Date	Version	Notes
March 18, 2026	v12.5.2.1	Some urgent requests/fixes and updating DEF core code.
January 26, 2026	v12.5.2	Major release with AH integration, support for the v11 ASR, and a few other bug fixes/enhancements.
July 18, 2025	v12.5.1	Major release with many new features, enhancements, and bug fixes.
February 28, 2025	v12.5.0.2	Minor release with urgent fixes
January 31, 2025	v12.5.0.1	Minor release with enhancements to support pre-ingests and other customer requests
December 20, 2024	v12.5	Major release with many new features, enhancements, and bug fixes

Release v12.5.2.1 (March 18, 2026)

Enhancements

Improved the speed of the Workflow Enlighten action

The Workflow Enlighten action used to be a single-threaded that was taking longer time to process the documents. It is now multi-threaded, so this action can process the documents much faster.

Remove the default Email address from the NEAT Notification

There was a default Email address configured for the “@nexidia.com” domain. Not all the customers use that email address and were sending emails to the wrong location. We removed this default value, so the users can enter whatever email address makes sense for that customer.

Support to ingest the data that is related to transferring to virtual agents

The DEF used to remove the virtual agent data from the transcript and send data only between the human agent and the customers. DEF can now ingest data that was part of the transfer between the virtual agents and customers.

DEF returns the proper error message when JSON doesn't return any participants

While retrieving call recording metadata, DEF now returns an error message when the metadata JSON doesn't return any participant information, so the end user can fix the data issues.

Mandatory to use the LanguageID configuration in DEF action

If the LanguageID field is not configured in DEF's Convert Transcript To NxPr action, DEF doesn't utilize the STOP word filter. Then, DEF will not send the best phrases to the NIA because it doesn't recognize the STOP words. DEF now forces the user to select the value in the LanguageID field, so the action/process will not work without this configuration.

Enhance the way DEF handles the complex LLM prompt

DEF sends most user-written prompts just fine to the external LLMs. However, if the prompt is very complex, written using a few special characters, it used to throw an error message. DEF now formats the prompt that is compatible with the LLM before sending it to the LLM.

DEF will calculate sentiment by default rather than NIA

The sentiment calculation can be done at two places, DEF and NIA. We updated the DEF configuration "Include Sentiment" to True in the Workflow Enlighten action, so the sentiment calculation will always happen in the DEF going forward, so we will not calculate sentiment scores in the NIA.

New Actions / Features

Extract the complaints data from TAS 5.5 and send them to the NIA

This capability allows for the identification and extraction of interactions from the complaints data, and then sends them to the NIA platform to make them accessible to customers. This data will be in the same format as intent data, in a hierarchical L1, L2, and L3 format.

Google Metadata Media Retrieve Action - Integration with Google API to retrieve media and metadata

DEF can now integrate with Google API and retrieve media and metadata successfully.

Generate Desktop Tags Action - Supports ingesting 3rd party evaluation and converting them into desktop tags

DEF allows for the ingestion of evaluations done by the 3rd party vendor and evaluations generated by QC and converts them into desktop tags before ingesting them into the NIA. So, they can be searched and reported by Explore and Reporting applications respectively.

Merge two DEF actions into one action

DEF has two separate actions: LLM Summary and JSON Creator. These actions must always work together. Keeping them separate creates extra work in the process creation. Thus, we decided to merge both functionalities into one action. The new action Auto Summary LLM v2 can be used to perform the same actions that were done separately.

Key Resolved Issues

ID	Description
NXQC-65569	UI cannot be used to apply Patch or Custom Action
NXQC-65489	DEF - Inbox Publisher configuration of the maximum number of files does not work as expected.

Release v12.5.2 (January 26, 2026)

General Updates

This release includes new features around Analytics Hub integration, support for the v11 ASR, and a few other bug fixes/enhancements.

Warning

If we want customers to use v11 and related features, DEF 12.5.2 is a must. These features will not work without DEF 12.5.2.

New Features

Ability to integrate DEF with the v11 VoiceGrid.

The DEF is used to integrate with the VG, which has a Workbench & workflow tool for ASR. We have a new solution where VG supports new v11 ASR and will reside on AWS. This capability enables the DEF to integrate with the v11-based VoiceGrid in AWS, performing ASR and completing the remaining processing as usual after obtaining the transcript from the v11 VG.

Ability to integrate DEF with the v11 ASR engine in Analytics Hub.

A new v11 ASR engine is available in the Analytics Hub (AH) in AWS, which will generate more accurate transcripts. The DEF starts using that tool to transcribe the media files. The DEF can integrate with the AH to create a transcript using the v11 engine, utilizing the Analytics Hub Request and Analytics Hub Response actions, and continue processing as usual with the other DEF actions.

Supports parsing metadata from transcripts generated by real-time ElevateAI.

ElevateAI can generate a real-time transcript and publish it on the message broker. The DEF can ingest the transcript immediately after the call is finished for further processing with the available metadata and send this data, including the transcript and partial metadata, to the NIA. Then, DEF will receive the remaining metadata and media files later as part of the post-call ingestion and merge all this data into the NIA.

Retrieves the IPM model file from the AH to generate Topic AI data.

The DEF integrates with the Analytics Hub and can retrieve the IPM model generated by the Topic AI Editor, configuring it into the DEF action. This will help generate Topic AI data using the updated model, eliminating the need for manual updates to the new IPM model in the DEF action.

Rename two DEF actions to Topic AI Request and Topic AI Response.

We have separated the AutoSummary and Topic AI data recently because both use different approaches to process interactions and produce data. The NiCE models produce Topic AI data, while external LLMs produce AutoSummary data. So, we rename two DEF actions, Call Summarization Request and Call Summarization Response, to Topic AI Request and Topic AI Response, respectively.

Adds the ability to write to an S3 resource with and without a KMS key.

The DEF writes to the S3 resource using a KMS key and without using a KMS key.

Supports converting the v11 ASR transcript to ASR format.

The Convert Transcript To ASR action can convert a transcript from the v11/AH format to the ASR format, allowing Enlighten models to generate scores using that transcript.

Added support to generate ENL scores for the v11 transcripts.

The DEF added support to convert the v11 transcript to ASR format via Convert Transcript to ASR, so we can generate the ENL scores using it.

Updated the way DEF calculates the best phrases.

The DEF now calculates the best phrases in the same way the Discovery service does. The DEF removes the stop words from the transcripts.

Installation/CU Updates and Enhancements

An ability to view the history of the patches that have been applied.

The DEF can now save and display the history of patches. It can show when the patch was applied, what the patch contained, and who applied it.

DEF New Actions and Resources

AWS SQS resource

The DEF communicates with the Analytics Hub using the AWS SQS resource to send and receive messages from the AH.

Analytics Hub Request

This action will push media to an S3 bucket in Analytics Hub, and also push messages to the SQS queue.

Analytics Hub Response:

This action retrieves messages from another SQS queue and uses the S3 path in the message to retrieve and store the transcript, converting it internally to VG ASR format before writing to the file system. It will pull messages in batches of 1,000 (configurable) and multi-thread their retrieval from S3.

Elevate AI Near Real-Time Retriever

ElevateAI can generate a real-time transcript and publish it on the message broker. The DEF can ingest the transcript immediately after the call is finished using the new action Elevate AI Near Real-Time Retriever for further processing with the available metadata and send this data, including the transcript and partial metadata, to the NIA. Then, DEF will receive the remaining metadata and media files later as part of the post-call ingestion and merge all this data into the NIA.

Compatibility Matrix

Application	Supported Version	Recommended Version	Notes
OS	2019	2022	
Database	2019	2022	
NiCE Interactions Analytics (NIA)	12.5	12.5 FP2	
NxDeployer	12.5.200	12.5.200 & above	
Language Pack	7.0 – 9.3	7.0 – 9.3	
Phase Pack	8.8 – 10.11+	v10.12	
Voice Grid	20.0.0	20.2.2.1	
Real-Time Grid	12.5.1	12.5.2	
Engage	7.x	7.5	

Key Resolved Issues

ID	Description
NXQC-61629	CEP timestamp not getting updated
NXQC-60662	SentimentTransition is incorrect format
NXQC-60992	NxInteractionUID may change on updates
NXQC-62494	CallSummaryResponse: accept NULL in Enlighten data
NXQC-61018	SENTIMENTTRANSITIONTYPE_NONE should be an integer

Known Issues/Behaviors and Workarounds

Issue	Workaround
When the File Mover Action is configured as: <i>Process from metadata=true, File Path Column name=ExternalMediaId</i> (for example, anything other than FileName column), and <i>flatten output directory=true</i>, the output files are still nested and are not flattened.	On the first File Mover Action set update <i>file key upon move=true</i>. Add another File Mover Action and configure it as follows: <i>Process from metadata=true</i> <i>File path column name=FileName</i> <i>Flatten output directory structure=true</i>
The new Maintenance Mode feature does not prevent attempts to publish Hierarchy data to Nexida/QC.	While a Nexidia inbox is not available, attempts to publish Hierarchy data will fail and retry; when the inbox is available again, DEF will catch up and publish data on the next retry.
After the Java 11 upgrade, it will show "An illegal reflective access operation has occurred" WARNING message.	This is a warning and can be disregarded. We will resolve it in the future DEF release.
If SFTP is selected (during DEF installation) as the means to connect NEAT & NEST, then in the installed system, the SFTP connection between NEAT & NEST cannot be made. Since NEAT and NEST cannot communicate, the DEF UI is non-functional, so DEF cannot be controlled or monitored via the UI.	Select a means of NEAT/NEST communication other than SFTP

Release v12.5.1 (July 18, 2025)

General Updates

This release includes many new features, enhancements, a few integrations, and bug fixes.

Warning

- This DEF release, 12.5.1, works with the Java 21. If you use an external JRE, please upgrade it to Java 21 before starting the new DEF installation or upgrade process. Please refer to the DEF installation document.

New Features

Provide the ability to decrypt the password stored in the DEF action.

Once the user enters the password to any DEF resources, it can't be seen for security reasons. However, this causes problems while troubleshooting the issue because the users don't know what password was stored for those resources. DEF now allows users with specific roles to decrypt the password and audit log of that activity.

Provide the ability to automate key rotation without the Assumed Roles.

This feature is similar to allowing the rotation of keys with the assumed roles. This will allow the users to do the same operation without the Assumed Roles, which is to prevent rotating keys manually. The DEF allows the rotation of keys in an automated way to improve productivity and supportability. This change is part of the **Amazon S3 resource**.

Ability to differentiate various DEF instances for monitoring purposes.

The DEF adds a new configurable property at the process level. This property allows the user to set values such as Production or Testing. The monitoring tools will monitor the DEF instances based on the value set in this property. This is a process-level configuration change, not an action-specific change.

Ability to collect some additional metrics for the chat media for the new ingest path.

There were still a few gaps in chat processing in the new ingest path that weren't considered when removing TextGrid. Stats like how long it takes the agent to reply to messages on average in the interaction, e.g., were stuff that TG did. If we remove TG, we won't be able to calculate these metrics, so we couldn't remove TG until we filled these gaps. This is handled via a **Media Statistics Calculator action**.

The DEF's enhanced logs show information from only one process at a time.

The DEF started showing logs from the other processes under enhanced logs, which were making troubleshooting extremely difficult. The DEF's enhanced logs now show information from one process only at a time.

Provide a mechanism to identify whether the CSV file is complete or not.

The DEF DB-to-CSV action will add an option to append NXTMP at the end of the CSV file name. For example, filename.CSV.NXTMP. This way, the process will be that the CSV file is not completed yet, and the system doesn't grab it while it is still in progress. When the file is complete, the action will change the name to filename .CSV.

Allows skipping the bad CSV file during DEF processing.

Currently, if a 'bad' (formatting error) CSV file is fed to the **CSV-to-DB action**, the action errors out and causes the process to abort. Now, when the process runs the next time, the same CSV file will be tried again, leading to the same failure. That would happen until someone manually resolves it. Instead, DEF now provides an option to skip a bad CSV. The default behavior would be the current behavior, which is 'do not skip'.

Allows the option to rename fields to avoid conflicts.

There are scenarios where the fields retrieved by the CXone Generic Metadata Retriever may have the same name as other existing fields, e.g., StartTime. The DEF now has an option to allow a prefix to be added to field names.

The sensitive fields were removed from the Call Summarization Response.

A few sensitive fields, such as ExternalMediaId, TranscriptUri, etc., in the Call Summarization Response were showing up by default. The DEF now excludes those fields from that JSON file.

Enhancements to the "Cleanup Type" action for better folder management.

The "Cleanup Type" action now has three more options for better folder management. It will allow the user to either "Delete empty folders," which is a default option, or "Never delete empty folders," or "Delete empty folders after X hours."

Provide the ability to generate an ASR format transcript from emails.

The DEF now has a new option in the **"Parse Email Text to JSON"** action that will allow us to generate an ASR format transcript for emails so that VG can process it using the various models and generate relevant scores to the NIA as part of the new ingest path.

Provide the ability to generate an ASR format transcript from general text media types.

The DEF now has a new option in the "**Parse General Text to JSON**" action that will allow us to generate an ASR format transcript for the general texts so that VG can process it using the various models and generate relevant scores to the NIA as part of the new ingest path.

DEF shows a warning if there are mappings that are not in the working table.

When designing a new DEF process or modifying an existing one, the user typically performs the mapping between fields. If there is no column in the WT for a given field, it still maps it successfully, but this can cause a problem later. Instead, now, when the user clicks "Submit," DEF checks to see if the Working Table has a column for a specific field. If it doesn't, it shows a warning so he can fix this mapping immediately.

Allow Process Launch and Process Chain actions to refer to the launched process via ID.

The Process Launch Action or the Process Chain Action is used for Process A to launch Process B, and it is referred to by its name. If a person changes B's name, the corresponding Launch/Chain Action must be updated. To resolve this, these DEF actions now have an option of referring to B by its Process ID.

Need to empty the key fields while importing/exporting processes.

When DEF exports/imports processes to use them in the other environment, it keeps the information in access/secret key fields, which can cause problems for another environment or the customer's process. DEF now empties access/secret key fields to prevent this problem from happening.

Associate the right speaker with the right channel.

When the outbound calls have a different channel assigned to the speakers than what NIA expects by default, the CXone Media Retriever action grabs channel allocation details from the playback API when getting the media. The DEF's Workflow Response action provides an option to flip the channels if the speakers are not on the channel that NIA expects.

Add information such as scheduling, threads, and resources to the DEF Templates.

The DEF doesn't add additional information to the DEF template/exported processes, such as scheduling, threads, resources, and a few other parameters. DEF processes will have this information while exporting/importing them, so the user doesn't need to enter it manually.

Enhanced CXone Media Retriever action to support IVR Data.

The **CXone Media Retriever** action can now extract IVR data. This will help to recognize and process the IVR channel type, similar to how it handles the Voice channel type.

Support for the SOCKS5 Proxy was added to the SFTP resource.

The DEF now allows a new configuration option, "Enable SOCKS5 Proxy," to the **SFTP resources** with True/False options. The default is set to False. When enabled, it displays additional configuration fields for Proxy Host and Proxy Port, both of which are required with no default values.

Pass the "SentimentTransition" value from the VG to the NIA.

There are a couple of sentiment values, such as OneHalfSentimentScore, ThreeFourthsSentimentScore, and SentimentTransition that are not being passed to the NIA. The DEF will calculate and/or retrieve these values from the VG and send them to the NIA.

Provide a separate retention period at the process level.

When we want the working table of a specific process to have a different retention period than at the server level, it requires performing multiple steps. The DEF now provides an option to set retention, in days, at the process level that will override the Server Retention value.

The CXone Users Retriever v2 Action can return "hireDate."

The CXone Users Retriever v2 Action is enhanced to return "hireDate" for the user's default search filter.

Optimize the AppLink Metadata Retriever to process a large number of data.

If the AppLink Metadata Retriever gets a huge number of calls to process, it can take a very long time to make it through them all. Now, DEF can process a very large amount of data in a "short" time.

The CXone Metadata Retriever skips records with a null ACD ID.

Currently, if the CXone Metadata Retriever gets a null value for ACD ID, an error is logged, and the process exits. The DEF can now continue running and log these records with the Nxfailure message.

DEF captures additional info in the Process snapshots.

In a DEF Process snapshot, only Actions are captured currently that support history tracking. DEF now provides the same support for history tracking for the Resource components.

DEF adds the middle name to the user details CSV.

The DEF adds a middle name when creating a user CSV file from the Engage extracted data.

Provide the ability to redact the email media type.

The DEF now provides a redaction capability to the email media in the **Parse Email Text to JSON action** while converting them to TextGrid format and before sending them to the NIA.

Sends a few missing data to the NIA first-class fields for the Email media types.

The DEF sends a few missing fields to the NIA first-class fields for the Email media type. This change is added to two DEF actions - **Parse Email Text To JSON** and **EML Parse Text**.

Provides the ability to trace the interaction E2E between DEF and NIA.

There is no way to trace the interaction E2E from DEF to the NIA, which creates challenges while generating reports. The DEF has a new working table column called "NxInteractionUID" (a GUID) that will be used to trace the interaction E2E. This information will be stored in the NIA first-class field with the same name. The inbox publisher will show this field as the default mapping.

Simplify the failed interactions re-processing task.

When the interactions fail in the DEF for any reason, it takes more knowledge and steps to reprocess them after fixing the problem. The DEF created two new tables to simplify the reprocessing in a few steps.

A few enhancements to the AutoSummary feature. All these changes are part of the Call Summarization Response v2 action.

Intents and Events can be visible on the player's waveform at a specific time.

The DEF adds offset values for the intents and events data that is generated by the **Call Summarization Response v2** action before sending them to the NIA. The user can see this information on the player's waveform at a specific time.

Removed the rule-based summary.

There will be only an LLM-based summary in the Nexidia Analytics platform. The rule-based summary did not add any value to the agents on top of the Topic AI data.

Sends failed interactions information to the NIA that can't be retried.

There are a few scenarios where the VG will send a few failed interactions because it doesn't have intent data. The DEF currently keeps retrying them until it gets resolved, but that is not a good approach because those interactions carry important information needed by the NIA, and the retry approach will not fix those interactions. DEF will now not retry those interactions and will send them to the NIA with any additional information it may have.

Installation/CU updates & enhancements

Updated the configuration file to work with the NxDeployer to automate the DEF upgrade process.

The DEF creates a configuration file, batchCUTemplate.properties, that the NxDeployer will use to upgrade the DEF in an automated way on multiple servers without human intervention. This will eliminate the need to enter information manually in the NxDeployer UI and future development whenever the DEF adds/updates configuration.

Move the Workbench folder from the data folder into the install folder.

The issue is that DEF is reading Workbench binaries out of the DEF data folder, which typically goes on the media drive, which is encrypted, and that runs counter to the company's IT standard practice of keeping binaries on the C: drive (in the install folder). This practice causes performance issues because we have to decrypt all the data from the media folder. We would not have performance issues when the Workbench is in the Install directory.

Updated the support for Java version from 17 to 21.

DEF 12.5.1 runs on Java/JRE 21, which will provide better security and many other features that come with Java 21.

Support for Windows Server 2022.

DEF formally supports Windows Server 2022.

DEF New Actions and Resources

All configuration options for new and updated Actions and Resource Types can be found in the DEF User Guide.

Allow the extraction of DFO emails from the CXone for the Nexidia Analytics.

The DEF allows the extraction of DFO Emails from the CXone and sending them to the Nexidia platform for Analytics purposes. This can be accomplished using the new action called **CXone Completed Contacts Metadata Retriever v2**.

Ability to retrieve data using Zoom API.

The DEF allows for the extraction of media and metadata using Zoom API and sends them to the Nexidia platform to perform analytics on them. The user can use **Media URL Download** action to use this capability.

Provide the ability to feed billing data to the CXone billing mechanism.

The Nexidia team sends the billing data for the CXone customers who use the NIA platform or the XO application manually every week via an Excel spreadsheet. We automate this process using the DEF that will get the XO billing data from its DB and will send it to the CXone message broker. This can be accomplished using the new action called **CXone Billing Data Submit v1**.

Added a new resource type - Snowflake

As part of the CXone/Snowflake integration, the DEF added **a new resource type called Snowflake**. Only DEF's CXone Snowflake Extractor action will use this resource to extract the data from Snowflake based on the query retrieved from Artemis.

Added a new resource type - Artemis

As part of the CXone/Snowflake integration, the DEF added **a new resource type called Artemis**. Both new DEF Snowflake actions, CXone Snowflake Query and CXone Snowflake Extractor, will use this resource, first to retrieve the query from Artemis and then pull data from Snowflake.

Allows the submission of a query to Snowflake via the Artemis message broker.

This new DEF action, **CXone Snowflake Query**, will submit a query to the Artemis message broker that can be used to execute against Snowflake to retrieve data.

Allows the extraction of metadata from Snowflake using a query from a message broker.

This new DEF action, **CXone Snowflake Extractor**, will retrieve the query from Artemis and execute it against Snowflake, utilizing the Snowflake resource to extract relevant data from Snowflake.

Compatibility

Nexidia Analytics*	v12.3 and above
NICE Quality Central	12.2 SP4+
Voice Grid	• Redaction: Voice Grid 2.0 with Redaction Workflow Model • NICE ENLIGHTEN: Voice Grid 2.0
Sentinel	6.12 + HF534055
Engage	v6.12 and above
Supported Browsers	Chrome (v84+) and Edge (v84+)
Java	JDK & JRE - zulu21.36.18-sa-fx-jdk21.0.4-win_x64

Key Resolved Issues

ID	Description
NXQC-33720	Enhanced logs for one process contain data from other processes running at the same time.
NXQC-54960	CXone Generic Metadata Retriever: embedded map responses are processed by the generic metadata retriever.
NXQC-56657	Spaces in column headers causes the CSVToDBAction to fail.
NXQC-42487	DEF - CXoneTimeoutError (intermittent timeout errors in our action).
NXQC-56845	Issue with Negative Quantum for the DST date.
NXQC-55443	File filtering is case sensitive, but it should not be.
NXQC-57979	Remove VARCHAR from the index and value names; autosummarization tag, workflow enlighten
NXQC-57144	Issue with retrieving media stored on two AIR loggers.
NXQC-55491	Calabrio Media Retriever sometimes gets partial calls.
NXQC-55490	RabbitMQ interface to VoiceGrid can close unexpectedly.
NXQC-55169	Stylesheet mapper: Exported info can't be imported
NXQC-48846	EDP action throws an error message "divided by zero" upon entering 0 values in the configuration.

Known Issues/Behaviors and Workarounds

Issue	Workaround
When the File Mover Action is configured as: <i>Process from metadata=true, File Path Column name=ExternalMediaId</i> (for example, anything other than FileName column), and <i>flatten output directory=true</i>, the output files are still nested and are not flattened.	On the first File Mover Action set update <i>file key upon move=true</i>. Add another File Mover Action and configure it as follows: <i>Process from metadata=true</i> <i>File path column name=FileName</i> <i>Flatten output directory structure=true</i>
The new Maintenance Mode feature does not prevent attempts to publish Hierarchy data to Nexida/QC.	While a Nexidia inbox is not available, attempts to publish Hierarchy data will fail and retry; when the inbox is available again, DEF will catch up and publish data on the next retry.
After the Java 11 upgrade, it will show “An illegal reflective access operation has occurred” WARNING message.	This is a warning and can be disregarded. We will resolve it in the future DEF release.
If SFTP is selected (during DEF installation) as the means to connect NEAT & NEST, then in the installed system, the SFTP connection between NEAT & NEST cannot be made. Since NEAT and NEST cannot communicate, the DEF UI is non-functional, so DEF cannot be controlled or monitored via the UI.	Select a means of NEAT/NEST communication other than SFTP

Release v12.5.0.2 (February 28, 2025)

General Updates

This release has a few urgent fixes that have an impact on migrating existing customers from the current NIA implementation (post-ingest) to the new-ingest/all ASR approach.

Create Workflow Enlighten for SentimentValue and First Class Field

The NIA applications don't show the values of the SentimentValue field because the DEF doesn't send it to the correct first-class field. The DEF now sends this field information to the correct NIA first-class field. This should help to display this information in the NIA application.

DEF's Inbox Publisher test mode supports the Extractor metadata format.

The Extractor metadata is expressed in a different format than DEF expresses it. So, the DEF's comparison tool can't do a comparison. The DEF now takes the different metadata formats into account during the comparison.

DEF's Inbox Publisher test mode supports re-processing.

The DEF's Inbox test compare tool supports the new interaction, so the users can't verify the already ingested interactions. The DEF now supports the already ingested interactions as part of the re-processing.

DEF shows the older release name on the login page.

The DEF 12.5 release shows the older release name on the login page. The DEF will show the current release name as part of this fix.

Send the Sentiment value information to the NIA's first-class field.

The DEF was used to send the Sentiment value information to the NIA via AutoSummarization Tags, which is why it was not displayed in the NIA application. The DEF will now send that information to the NIA's first-class field, SentimentValue, so the user can view it in the NIA application.

DEF's release artifact's Zip filenames reflect release name.

The For the minor DEF release, the Zip files produces by the build process were not reflecting the same name as the release name, for example DEF 12.5.0.2 vs. DEF 12.5. This

works fine for major releases such as DEF 12.5 and 12.5.1. The build artifacts will show the same name as the release name for the minor DEF release, too.

Enhancements to the DFO chat integration.

- Added a new configuration to prefix all working table column names with "wt_" when the "Prefix metadata" option is set to true. By doing this, we will avoid having column name collisions in the output CSV file. Similarly, a prefix may be added to the DFO custom metadata column names, e.g. "cust_"
- The CXone Media Retriever V3 Action now supports the survey types of chat data. It also creates the CSV file with the same naming convention as the bot files. Also, the WT column name is updated to "ChatFileName."
- The DFO chat CSV file "DFOType" column is now initialized with the working table "ChannelType" column name.

Key Resolved Issues

ID	Description
NXQC - 55491	Calabrio Media Retriever sometimes gets partial calls
NXQC - 55490	DEF's RabbitMQ interface to VoiceGrid can close unexpectedly
NXQC - 55443	File filtering is case sensitive but it should not be
NXQC - 55169	Stylesheet mapper: Exported info can't be imported

Release v12.5.0.1 (January 31, 2025)

General Updates

This release has enhancements and new features to help customers migrate to the pre-ingest path and other customer promises.

Enhancements to the AutoSummary features.

- The DEF used to send the intent data and/or part of the transcript to the LLMs to generate the call summary. This approach negatively impacts its accuracy. The DEF now sends the entire transcript via **Auto Summary LLM action** all the time to improve the accuracy of the generated summary, except when it hits the token limit.
- The way DEF calculates the start and end time offset for the chat media was causing an issue. We made changes in the DEF based on the field request that caused the start and end time offsets to become the same. To resolve this, the DEF ensures via **Chat CSV to JSON V2 action** that the end time is greater than the start time in the CSV that will be used to create a JSON file.

Enhancements to Salesforce chat integration to remove chatbot data.

The DEF can now remove chatbot turns and customer responses to chatbots from the Salesforce action called **Salesforce Metadata Parser Action**. There will be two new options in the DEF's Salesforce action. First, Split Agent takes a chat conversation involving a customer and multiple agents and splits it into multiple chats, one agent per chat. For this operation, chatbots will be treated as "agents." Second, Strip Bot removes the chatbot content, i.e., chatbot turns and people's responses to chatbots.

Enhancements to DFO chat integration.

The DEF can remove chatbot turns and customer responses to chatbots from the DFO integration. There will be two new options in the DEF's **CXone Media Retriever V3 Action**. First, Split Agent takes a chat conversation involving a customer and multiple agents and splits it into multiple chats, one agent per chat. For this operation, chatbots will be treated as "agents." Second, Strip Bot removes the chatbot content, i.e., chatbot turns and people's responses to chatbots.

Help migrate existing customers to Pre-ingest ASR.

- The DEF creates a 'fake inbox' using its database space so that the users can verify the DEF-generated data to make sure they match with the NIA. This change is part of the **Inbox Publisher** action.
- Once the user compares and verifies the DEF-generated and NIA data, we don't want to keep the test object in the customers' production environment. The DEF allows the deletion of these "fake inboxes." This change is part of the **Inbox Publisher** action.
- The DEF has a tool that will take an Extractor stylesheet definition and define a corresponding DEF inbox mapping. This tool will help expedite the customer migration from the Extractor to the DEF. The user can do this using the **Convert Xls To Metadata Mapping Action**.

Provide an ability to retrieve data using Zoom API.

The DEF will extract media and metadata from Zoom using the new action **Media Download URL Action**. As usual, the user has to select the right options from its various configurations based on the customer's requirements.

Allows to extract DFO Emails from the CXone APIs.

The DEF's action "**CXone Completed Contacts Metadata Retriever V2Action**" can extract DFO emails from the CXOne API and send them to the NIA for Analytics purposes. This action will use the value of the "updatedSince" attribute to pull the emails. It will pull 1000 by default, which can be configurable if needed.

Enhancing Twilio API integration action.

The current DEF action, **Twilio Media Retriever Action**, for the Twilio integration, extracts mono recordings for analytics purposes. This action will now extract stereo recordings for the NIA using new APIs.

Collects additional metrics for the chat media for the new ingest path.

The DEF creates a few metrics for the chat media using the "**MediaStatics Calculator**" action. We found that there were a couple of metrics missing in the current implementation. The DEF creates a few more metrics for the chat media as part of this change, such as StartTimeUtc, EndTimeUtc, TurnCount, etc.

Release v12.5 (December 20, 2024)

General Updates

We have introduced many new features including new actions, many enhancements and a few bug fixes in this release.

Breaking Changes

- **For the customers who want to use TLS 1.3** - When DEF is already installed, they can enable strict encryption on their SQL Server instance, then add the server certificate to DEF's certificate store and configure DEF's `ndef.properties` to connect to SQL via strict encryption. However, when DEF is a clean install (first-time installation), they need to disable strict encryption in SQL Server, install DEF as normal, then re-enable strict encryption in SQL and configure DEF accordingly.

We bundle the JDBC driver with DEF. It gets installed/upgraded when DEF is installed/upgraded. While older JDBC drivers bundled with older DEF should be able to connect to SQL Server 2022, it may be a best practice to use a newer DEF with SQL 2022 since modern DEF contains newer JDBC drivers certified for SQL 2022.

- **Removal of the Silent Installer configuration** - We haven't updated the Silent Installer functionality for a while. The reason is that we keep `MultiInstall` and `MultiCU` configuration files & their properties up to date, which provides the same functionality as Silent installer, which is to automate DEF deployment. This makes the Silent installer feature redundant and unnecessary, so we removed it from the 12.5 release.
- **Updating Self-signed certificate** - The DEF's current self-signed certificate that allows access to NEAT UI over HTTPS is going to expire in October 2025. When that happens and whichever customers haven't updated the self-signed certificate with the authority-signed certificate, they will see browser warning signs or even cause problems in accessing DEF UI if their IT security policy is set up in a certain way. To resolve this, the customer has to generate a new certificate by themselves or with the help of Nexidia support and upload it to their DEF, or they will have to upgrade the DEF to 12.5 when it is released in December.

New Features

Updated DEF's self-signed certificate that allows to access NEAT UI over HTTPS.

The DEF's current self-signed certificate that allows access to NEAT UI over HTTPS is going to expire in October 2025. When that happens, if the customer hasn't updated the self-signed certificate that ships with DEF with an authority-signed certificate, they will see browser warning signs, or may even have problems accessing the DEF UI if their IT security policy is set up in a certain way. The DEF got updated with the self-signed certificate for another decade.

Ability to support different intent models in the new VG cluster arrangement.

The DEF supports multiple models based on the different metadata so long as those multiple models are deployed on the same VG server. However, the IT team may decide to use only one model per VG server. That means there will be multiple VG servers for multiple models. The DEF now supports this deployment use case, too.

Provide an ability to hide/show DEF features based on certain configurations.

The DEF has a few features that are not applicable for certain users. We added a mechanism that would allow us to hide/show features from the user that are not applicable to them.

Provide an ability to select media based on the metadata selection.

The DEF does not allow the user to choose specific media from the S3 resources. So, the user has to pull all the media whether they need them or not. This feature is available in all other resources. The user can now choose specific media based on the metadata selection for the S3 resource, too. This will avoid using a lot of space, reduce cost, and improve performance.

Make the deployment cluster-aware using the heartbeat.

There was an issue in the production where multiple DEF instances were talking to the same DEF DB, which caused a problem. To prevent this from happening again, each DEF instance publishes a "heartbeat" into the database on a configurable interval (default = 60s) and also checks the database on that same interval to see if more than one such heartbeat is coming in. If so, it must shut itself down, and fire an error message that goes to an email indicating that a cluster misconfiguration has triggered service shutdown.

Provides the IPM information to the NIA to retrieve them later for other use.

The DEF uses the IPM file to generate the intent data. The DEF sends this intent data and IPM file information to the NIA. This information later can be retrieved, for example, by XO, to know which IPM file was used to generate this intent data, so they know which IPM file to use in the XO and not duplicate the effort by using the same IPM file that the AutoSummary used.

Updating the DEF monitoring data to support UTC format.

The NIA DB uses the UTC format for the monitoring data. The DEF updated its monitoring data to UTC format to match with the NIA to prevent issues that may occur because of the different time formats between the two systems.

Provide an ability to recalculate the sentiment and ENL scores without re-ASR.

If we need to calculate the ENL scores using the VG for the post-ingest interactions, it can't be done right now. To address this need, the DEF will pull post-ingest NxPr files, convert them to VG ASR transcript form, feed them back through Workflow Request/Response/Enlighten to get fresh Enlighten-only scores, and inboxes Enlighten + Sentiment back to NIA as partial metadata update.

Update the CXone Metadata Retriever action based on the CXone API changes.

The CXone Metadata Retriever action was updated to include a timestamp besides specific hours in a day to retrieve metadata from the CXOne API. This way, the user can retrieve CXone data multiple times a day. The DEF also made changes to how a few business data are getting processed in the backend, which will reduce some work by the SAs.

A couple of minor enhancements were added to the rule-based summary.

The DEF updated the rule-based summary to cover invalid scenarios, such as when the ENL scores are out of range. Also, the DEF updated the wording of the rule-based summary for better readability purposes. For example, it used to show Yes & No for the Outcomes. Instead, it reads "is Resolved or Not Resolved."

DEF now provides the top 3 intents of the interactions.

The DEF is used to send the primary intent only for the interactions to the NIA and AS JSON files. The DEF now provides the interactions' top 3 intents, if available. So the customer can understand the interactions with multiple intents better.

Extract a few more fields from the Engage for the AppLink.

The customer wants data from 4 more fields from the Engage DB. The DEF now retrieves data from these fields - Customer Identity, Customer Enrolment Status, Authentication Status & Score, A Call Enrolment Status.

Add "publish confirmations" to both DEF and VG clients to confirm that processing is done.

There are situations where the AS system loses the RabbitMQ messages (mainly when the server restarts or the media server is unavailable). To prevent this from happening, the DEF adds publish confirmations to both DEF and VG clients when they push to Request/Response queues to make sure RMQ responds that it's done before the DEF calls the push done. This ensures that the request is written to disk, guaranteeing that the DEF won't lose any messages pushed to the queue.

Enhance Enlighten scores functionality to be compatible with ElevateAI integration.

To tighten DEF-Elevate AI integration for multi-tenancy deployment and remove VG from the deployment, the DEF sends the necessary data to ElevateAI to generate the ENL scores precisely the same as the VG generates.

Provide an ability to automate key rotation with the Assumed Roles.

For the DEF's integration with Amazon S3, we have to rotate the keys manually every couple of months. This is inefficient and can cause problems in production if the users forget to rotate the keys on time. The DEF allows the rotation of keys in an automated way to improve productivity and supportability.

Ability to use a specific intent model based on the interactions metadata.

All interactions use the same intent model to generate the intent data. This may not generate more accurate data about the interactions from the different LoBs, languages, etc. The DEF now

allows configuring different intent models by selecting specific metadata. This will help generate more accurate intent data based on the interactions.

Allow users to resize windows in the DEF UI using a mouse.

The DEF updated the scroll bar behavior based on the services team's request. The inner scroll bar allows you to scroll the inner list of configuration properties. The outer scroll bar allows you to scroll the entire modal content. This will allow you to scroll down to the "Submit" button.

Replace the custom SQL that generates the index scores for the AutoSummarization tag.

There is a customer SQL that creates the index scores for the AutoSummarization tag. This custom SQL is very brittle and difficult to templatize and it is hard to change for different customers. The DEF productized this functionality in the Workflow Enlighten action to make it easier to use and support the services team.

Ability to create behavior scores metadata in the Workflow Enlighten action.

The SAs have to use custom SQL to generate the behavior scores, which are fragile and prone to errors. The DEF added a table with the behavior data and generates the scores using the Workflow Enlighten action, eliminating the need for custom SQL.

Create _Thresholds columns for the Enlighten models when there are values in the threshold fields.

The DEF creates the _Thresholds columns in its Working Table only for the fields for which the user enters the threshold values for any model. Otherwise, it will not create the _Thresholds column.

Provide a way for the FileDiscovery to work with other columns too besides "FileName."

When the File-discovery action is set to "process from metadata", it only looks at the "FileName" column of the working-table to do the discovery. DEF now provides a way to use other columns also to do the discovery for a FileDiscovery action.

Deletes the "*.ASR.json" files after processing from the Call Summarization Request.

Call Summarization Request is not deleting the "*.ASR.json" files after processing. The user needs to use a separate "Cleanup" action to delete them, which is inefficient. The Call Summarization Request will delete these files automatically after processing is done.

DEF ensures chat durations are not zero.

The chats presented to DEF come with a "start time." The DEF calculates the "end time" based off the last chat turn. If the DEF is presented with a chat with a single turn, the "end time" is set to the same value as the "start time". When this is ingested to the NIA, the chat duration is end-start=0. That can interfere with the evaluation of a group of single-turn chats. The DEF now applies a constant # of ms, for example, 5 ms, to add to the hits and then add a new config option for the session's end time.

Added a min/max time option to resolve the issue around retrieving decryption keys.

The SAP metadata upload action had an issue related to the time window used to retrieve decryption keys. If there is a heavy load, not all keys may be retrieved. The AppLink Metadata retriever action has min and max configuration and it is based on each quantum. For the SAP upload action, the DEF now directly queries the working table with the min/max function to get the min and max values regardless of which quantum we are in.

Updated default query in AppLink Metadata Retriever to avoid purged calls.

The query that AppLink Metadata Retriever action currently uses can include purged calls. That returns a very large amount of unnecessary metadata to DEF, more than it can consume quickly. This causes a performance issue. The DEF uses the updated query that excludes purged calls to improve the performance.

Provides an ability to limits the number of events to be displayed.

The IPM model generates tons of events from the interactions. Most of them are not needed by the end users. The DEF provides a configuration to limit the number of events to make them available in the NIA or the JSON file.

Provides the ability to remove existing custom actions and better validations.

If the customer has a custom action installed in their system, and the DEF core code has that functionality, then the DEF will remove the custom action automatically. If the customer has multiple custom actions, the DEF upgrade process will show an error message for the customer to resolve the conflict. The DEF provides better validations for the installed custom actions with these changes.

Enhancements to the Enlighten processing.

If we need to do the Call Summarization for the existing implementations, we can only do it by updating their process flow. The Call Summarization action can take the NxPr files generated earlier during the ingest and then pass them through the Call Summarization action. This way, NIA can have Call Summarization results along with the partial metadata update.

Allow the use of newly created NIA first-class fields without upgrading the DEF.

Whenever the NIA team adds a new first-class field, the services team has to upgrade the DEF to the latest version to take advantage of that newly added first-class field, which is inefficient. The DEF updated its approach to handling newly added first-class field data so that the SAs don't need to upgrade DEF to the latest version to use that field.

Updated the default value for "Download voice and screen media" to false in the CXOne Media Retriever V2 action.

When the customers don't have screens to ingest and the default value for the field "download voice and screen media" is set to True, it causes problems when no screens are available and doesn't download audio either. It downloads the audio portion successfully by keeping the default value of the above field "false."

Ignore the pair when the property name is empty for CXOneGenericMetadataRetrieverV2.

When there is no value in the property name of the CXOneGenericMetadataRetrieverV2, it doesn't return any value, which causes confusion downstream. The DEF will ignore the entire pair in the future if the property name is empty.

Allows to enter negative values in the Quantum field.

The DEF does not allow negative values in the Quantum field, which can allow us to pull the historical calls but pull the latest data first and move backward. The DEF now allows negative values in the quantum field so processes can start retrieving data backward from a given point in time.

Provide the ability not to publish the copied process automatically.

The DEF fails silently when the user tries to schedule the process, which was copied from the other process which has not yet been published. The DEF now shows a message to the user if the copied process is not published. The DEF never publishes the process automatically after copying it.

Enhance the "Remove Duplicate Metadata" action.

In the current "Remove Duplicate Metadata" action's behavior, the FileName must be present in the rollover table for it to be removed from the working table. The DEF adds the ability to remove the metadata duplicate within the working table if the "OriginalFileName" is the same.

DEF allows an option to use HTTPS in agent/user communication to NIA.

When DEF sends agent/user information to NIA, it uses HTTP. DEF now supports HTTPS to send this information.

DEF displays the release name, e.g. 12.4.1FP2, rather than the release number, e.g. 12.4.1.152.

The DEF displays the release number, e.g., 12.4.1.152, which doesn't help much when the services team reports any production issue for the R&D team to reproduce. The DEF now displays the release name, which is easy to identify to install in the internal environment to reproduce & troubleshoot it.

Provide an ability to optionally check signing before decrypting in GPG Decrypt action.

The GPT Decrypt action doesn't check the signature when decrypting the file, which is an area of concern for users who have security as a top priority. The DEF added a signature check feature optionally for whoever wants to verify the signature of the files.

Provide the ability to choose either an interval or fixed execution schedule.

The DEF currently allows an interval to be selected for its process execution scheduling (every 2 hours, 30 minutes, 1 day, etc.), which isn't always ideal when customers want a process to always run at the same time of day irrespective of daylight savings time adjustments. The DEF now supports fixed time selection for its scheduling (03:00:00, 05:30:00, 18:45:00, e.g.) . This will give users more choices when selecting quantum based on their needs.

Provide the ability to skip bad emails and let the process continue to other media.

When bad emails come via EML Parse Text action, the whole process fails, so the system can't ingest new data anymore. The DEF now logs and skips the bad emails and moves forward, ingesting the next data without failing the process.

Enhance Genesys Cloud Job Metadata Retriever Action.

The DEF updated the code to handle the Genesys Cloud Job Retriever API responses 200 and 202 differently. It now gracefully handles possible "stream closed" exceptions.

Enhance dealing with "Forbidden" responses from Genesys Cloud Call and Recording Metadata Retriever Actions.

We found out that the "forbidden" responses from the API are related to individual call recordings/requested resources and not something that is being applied to the overall access to the API, which is considered "unauthorized." The DEF made changes to handle the occurrences of "unauthorized," "forbidden," and "rate-violation" responses differently.

Handles the intermittent issue with CTI data on the first of the month.

DEF's retrieval of CTI metadata works fine all the time for a given quantum window. However, when the quantum happens to cross a midnight boundary on the 1st of the month, something is wrong with the retrieved data. DEF then tries to get audio that doesn't really exist. The DEF now handles this scenario gracefully.

Created a utility package that can be patched alongside actions.

DEF has improved the patching process for better supportability, including allowing multiple, one-per-action patch files, as well as a new process for patching utility classes containing code shared across multiple actions. There are a few areas where we have to write duplicate code across some actions rather than the better practice of having shared helper methods that they all use. When we have to patch one of these common codes, we have to update them in all places and provide all the actions, even if it is a common fix. In these scenarios, we will have to patch only one utility class in the future.

Installation/CU updates & enhancements

Improve automated deployment of the DEF using the MultiInstall file.

The DEF's automated deployment using the MultiInstall file was not really automated because it kept showing a screen to the user to validate the information, which required someone to be present to approve at each step. The DEF eliminated the requirement for anyone to approve the configuration and allowed the deployment of DEF to be fully automated.

The default DEF ports are changed, and the user can edit them.

The DEF used the default port 8080 during its installation. DEF now uses new default ports: 30000 for NEAT HTTP, 30001 for NEAT HTTPS, and 30002 for all servers' shutdown hooks. The users should still be able to change the port number as they like.

Better management of the existing patches.

The DEF allowed one patch and a custom action to be included in the existing deployment. The DEF now allows multiple patches and action JARs in the system. Also, the reporting used to break after applying patches because of the updated column name in the working tables. The DEF will retain the existing column names in the future after applying the patches, which will not break the reporting.

Adds support for MSSQL 2022.

The DEF now supports MSSQL 2022.

Provide an ability to track custom actions better.

Currently, the CU tool allows the user to preserve a previously installed patch on the system being upgraded that caused some issues in the field. To prevent issues from happening because of this, the CU tool renames the existing patch, for example, `old_patchname.jar`, such that it can no longer be used as a patch except via subsequent manual operation.

DEF New Actions and Resources

All configuration options for new and updated Actions and Resource Types can be found in the DEF User Guide.

AutoSummary LLM

This action helps to generate interactions' summaries using LLMs. The DEF will integrate with the NIA's LLM service, which will connect with external LLMs such as ChatGPT and Mistral. The DEF will send the intent data and/or transcript to the LLMs via the LLM service, which will generate the summary and send it back to DEF, which will update the AS JSON and also send it to the NIA.

AppLink SAP Generic Upload

This new action allows the upload of reconciliation data and user data using the same action. The DEF has a new action SAP generic upload to allow uploading either reconciliation reports data or user information.

Call Summarization Request v2 AND Call Summarization Response v2

The DEF has implemented a different mechanism for configuring RabbitMQ for both Call Summarization request and response actions. We have removed the Queue name field from the Response v2 action and made the corresponding field on the Request v2 action mandatory. These actions also provide the top 3 intents, instead of just the primary earlier, as part of the AutoSummary data.

Calabrio User Metadata Retriever

This action provides an integration with the Calabrio API. It retrieves agents' information from its API and stores it in the DEF working table.

Salesforce Metadata Parser

The Salesforce Metadata Retriever action retrieves and stores the metadata in the DEF working table. Then, this action gets those metadata from the WT, parses them, and writes them into the output file. The output file would be the EML file for email data and the CSV file for chat data that is received as part of the API output.

Salesforce Metadata Retriever

This action helps integrate DEF with Salesforce to extract chat data. It will extract the chat data from the Salesforce API to analyze them in the Nexidia platform.

Convert Transcript to ASR

This action provides the ability to convert 3rd party transcripts to VoiceGrid ASR format. The DEF cannot generate ENL scores for the 3rd party transcripts because they are not in the ASR format for the VG to process. The DEF now converts the 3rd party transcripts into the ASR format for VG to process and generate the ENL scores.

Custom SQL V2

This action provides a mechanism to prevent basic human errors in the custom SQL. DEF's Custom SQL action was created to be supremely flexible for the SAs, but that causes some issues in production, especially around human errors. The DEF added a new version of the Custom SQL action, which will perform validation on the user-entered custom SQL and prevent issues in production.

Compatibility

Nexidia Analytics*	v12.3 and above
NICE Quality Central	12.2 SP4+
Voice Grid	Redaction: Voice Grid 2.0 with Redaction Workflow Model NICE ENLIGHTEN: Voice Grid 2.0
Sentinel	6.12 + HF534055
Engage	v6.12 and above
Supported Browsers	Chrome (v84+) and Edge (v84+)
Java	JDK - zulu11.54.24-sa-fx-jdk11.0.14-win_x64 JRE - zulu11.54.24-sa-fx-jre11.0.14-win_x64

Key Resolved Issues

ID	Description
NXQC-40667	DEF_Daylight Savings
NXQC-33716	The 'Metadata Import' action is mistakenly treating as a delimiter value
NXQC-43104	"Resolve SQL" button doesn't output correct values for a couple of fields with alias.
NXQC-49893	Can't connect to SQL using TLS 1.3
NXQC-48562	Change field name from SentimentScore to OverallSentimentScore
NXQC-45087	File filtering is case-sensitive but it should not be
NXQC-42211	CXone Media Retriever: Issue extracting just audio from audio + screen call
NXQC-38653	Creation of Parent folder in S3 bucket using S3 DEF resource
NXQC-24688	The processes are not dropping the temp tables thus the SQL sessions must still be remaining active after the process is complete.
NXQC-47487	The Decompress action sometimes fails on a zip file
NXQC-46986	Need to limit contact summaries
NXQC-46010	Insure that scores ingested to NIA do not have exponential format
NXQC-43875	Avoid another issue with library incorrectly parsing date string
NXQC-41761	Avoid unnecessary multiple requests per file
NXQC-41760	Avoid requests to recreate folders that already exist
NXQC-41758	Avoid unnecessary requests to GET BUCKET

Known Issues/Behaviors and Workarounds

Issue	Workaround
When the File Mover Action is configured as: <i>Process from metadata=true, File Path Column name=ExternalMediaId</i> (for example, anything other than FileName column), and <i>flatten output directory=true</i> , the output files are still nested and are not flattened.	On the first File Mover Action set update <i>file key upon move=true</i> . Add another File Mover Action and configure it as follows: <i>Process from metadata=true</i> <i>File path column name=FileName</i> <i>Flatten output directory structure=true</i>
The new Maintenance Mode feature does not prevent attempts to publish Hierarchy data to Nexida/QC.	While a Nexidia inbox is not available, attempts to publish Hierarchy data will fail and retry; when the inbox is available again, DEF will catch up and publish data on the next retry.
After the Java 11 upgrade, it will show “An illegal reflective access operation has occurred” WARNING message.	This is a warning and can be disregarded. We will resolve it in the future DEF release.
If SFTP is selected (during DEF installation) as the means to connect NEAT & NEST, then in the installed system, the SFTP connection between NEAT & NEST cannot be made. Since NEAT and NEST cannot communicate, the DEF UI is non-functional, so DEF cannot be controlled or monitored via the UI.	Select a means of NEAT/NEST communication other than SFTP